

Squeezing Theorem

1.6.4 THEOREM (The Squeezing Theorem) Let f , g , and h be functions satisfying

$$g(x) \leq f(x) \leq h(x)$$

for all x in some open interval containing the number c , with the possible exception that the inequalities need not hold at c . If g and h have the same limit as x approaches c , say

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$$

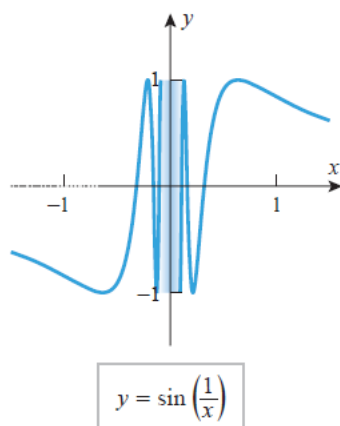
then f also has this limit as x approaches c , that is,

$$\lim_{x \rightarrow c} f(x) = L$$

► **Example 5** Discuss the limits

$$(a) \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \quad (b) \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$$

Solution (a). Let us view $1/x$ as an angle in radian measure. As $x \rightarrow 0^+$, the angle $1/x$ approaches $+\infty$, so the values of $\sin(1/x)$ keep oscillating between -1 and 1 without approaching a limit. Similarly, as $x \rightarrow 0^-$, the angle $1/x$ approaches $-\infty$, so again the values of $\sin(1/x)$ keep oscillating between -1 and 1 without approaching a limit. These conclusions are consistent with the graph shown in Figure 1.6.5. Note that the oscillations become more and more rapid as $x \rightarrow 0$ because $1/x$ increases (or decreases) more and more rapidly as x approaches 0 .



Solution (b). Since

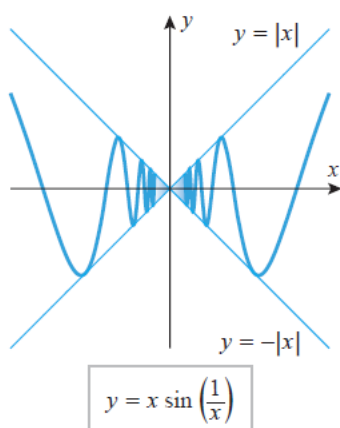
$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

it follows that if $x \neq 0$, then

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|$$

Since $|x| \rightarrow 0$ as $x \rightarrow 0$, the inequalities in (4) and the Squeezing Theorem imply that

$$\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$$



REMARK It follows from part (b) of this example that the function

$$f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous at $x = 0$, since the value of the function and the value of the limit are the same at 0.